

Q.18 Differentiate between elastic and plastic deformation. (CO2)

Q.19 Define Coupling. Explain its types. (CO5)

Q.20 What is Rankine's formula? Mention its limitations. (CO4)

Q.21 What are the inversions of single slider crank chain? (CO5)

Q.22 Explain elastic constants and derive relationship among them. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Derive the relation between Load, Stress, Strain, and Elastic Constants. (CO3)

Q.24 Draw SFD and BMD for a cantilever beam with a point load at the end. (CO3)

Q.25 With neat sketches, explain any two inversions of double slider crank chain. (CO5)

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6th Sem / Automation & Robotics

Subject : Solid Mechanics and Mechanisms

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Young's modulus is defined as the ratio of (CO1)

- a) Normal stress to normal strain
- b) Shear stress to shear strain
- c) Longitudinal strain to stress
- d) Volumetric stress to strain

Q.2 Malleable materials have (CO1)

- a) High hardness b) High plasticity
- c) Low ductility d) High brittleness

Q.3 In lower pair, there is (CO5)

- a) Surface contact b) A point contact
- c) A line contact d) Unpredictable

- Q.4 Which of the following is not a property of ductile material? (CO1)
- Yielding
 - Large plastic deformation
 - High tensile strength
 - Shattering without warning
- Q.5 In bending stress formula, the term $M/I = f/y$ holds good under (CO3)
- Pure torsion
 - Bending only
 - Shear only
 - All types of load
- Q.6 Whitworth quick return mechanism is used in (CO5)
- Lathes
 - Slotter
 - Drilling machine
 - None of these

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 A kinematic chain becomes a _____ when it provides a definite motion. (CO5)
- Q.8 The property of a material to regain its original shape after deformation is called _____. (CO1)

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- Q.9 Oldham's coupling is used to connect _____ shaft. (CO5)
- Q.10 Euler's formula is applicable to _____ columns. (CO4)
- Q.11 The value of section modulus for circular section is _____. (CO6)
- Q.12 Elastic limit is the maximum stress up to which material can return to its original shape without _____ deformation. (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What are the assumptions made in pure bending theory? (CO3)
- Q.14 Define modulus of rigidity and bulk modulus with units. (CO2)
- Q.15 Explain any two surface hardening processes. (CO4)
- Q.16 Draw shear force diagram for a cantilever beam with UDL over the entire length. (CO5)
- Q.17 Explain the working of Geneva mechanism with diagram. (CO5)

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